

ChE 330 – Polymer Science – Fall 2025

Student Learning Outcome 1

Rubric for assessment problem on HW2: Problem 3

Polymer engineering problem involving molecular weight averages, polymer fractionation, and polydispersity

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Does not correctly identify or apply the governing polymer molecular weight relationships. Shows limited understanding of how number-average and weight-average molecular weights relate to the polymer mixture system.	Identifies relevant polymer molecular weight relationships, but has gaps in applying them correctly to the fractional precipitation problem.	Correctly applies mathematical and scientific principles to formulate the polymer mixture model using number-average and weight-average molecular weight relationships.
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	Does not correctly set up or solve the algebraic relationships needed to determine the weight fractions or produces incorrect results due to major setup errors.	Shows partial understanding of the algebraic method but makes errors in substitution, manipulation, or calculation of the weight fractions.	Correctly uses algebraic relationships to solve for the weight fractions of polymer fractions A and B.
5. Predict and defend problem outcomes	Does not correctly predict the polydispersity index or does not connect the result to the physical meaning of the polymer mixture.	Shows partial understanding of the equal-polydispersity assumption but has errors in applying it or interpreting the final result.	Correctly predicts the polydispersity index and provides a reasonable engineering interpretation of the outcome under the equal-polydispersity assumption.

Averages:

Performance Indicators: (1) 2.35 (2) 2.35 (5) 1.68

Overall Average: 2.13

The strongest performance was observed for PI 1 and PI 2, both with an average score of 2.35. PI 1 assessed students' ability to apply mathematical and scientific principles to formulate the polymer mixture problem. Many students were able to recognize that the problem required molecular weight mixture relationships and could connect the given number-average molecular weights to the unknown weight fractions. This suggests that students are developing reasonable confidence in identifying the governing relationships for polymer molecular weight systems.

PI 2 also had an average score of 2.35. This indicator assessed students' ability to solve engineering problems using algebraic methods. Students who performed well were able to correctly substitute the given molecular weights into the number-average molecular weight relationship and solve for the weight fractions of fractions A

and B. Students who struggled generally made algebraic setup errors, confused number-average and weight-average relationships, or did not correctly enforce the condition that the weight fractions must sum to one.

The lower score was observed for PI 5, with an average of 1.68 and a relatively high standard deviation of 1.49. This indicator assessed students' ability to predict and defend problem outcomes, particularly by applying the assumption that fractions A and B have the same polydispersity. The lower score suggests that students had more difficulty moving from the calculated weight fractions to the interpretation of the polydispersity index. Some students could complete the weight-fraction calculation but struggled to connect the equal-polydispersity assumption to the weight-average molecular weight of the original polymer mixture.

Overall, the SO 1 average score was 2.13 with a standard deviation of 1.07. These results indicate that students generally met expectations for identifying, formulating, and solving the assessed polymer engineering problem. Student performance was stronger in formulation and algebraic solution than in prediction and interpretation of the final polymer outcome.

To improve student performance, future offerings of the course should include additional in-class examples that explicitly connect number-average molecular weight, weight-average molecular weight, weight fractions, and polydispersity index. More emphasis should be placed on helping students distinguish between the different molecular weight averages and understand when each relationship should be applied.

Additional practice problems will require students not only to calculate numerical answers but also to briefly explain the physical meaning of their results. In particular, students will be asked to justify whether calculated weight fractions and polydispersity values are reasonable for a polymer mixture. This will help strengthen students' ability to predict and defend problem outcomes rather than only perform calculations.

Future assessments may also separate the calculation and interpretation components more explicitly so that student performance on model formulation, algebraic solution, and engineering interpretation can be tracked more clearly across the course.

Expected Solution

For a polymer mixture, the number-average molecular weight relationship is:

$$\frac{1}{M_{n,P}} = \frac{w_A}{M_{n,A}} + \frac{w_B}{M_{n,B}}$$

with:

$$w_A + w_B = 1$$

Substituting the given values:

$$\frac{1}{178000} = \frac{w_A}{128000} + \frac{1 - w_A}{225000}$$

Solving gives:

$$w_A = 0.3484$$

$$w_B = 0.6516$$

Therefore, the weight fractions are:

$w_A = 0.348 \text{ or } 34.8\%$

$w_B = 0.652 \text{ or } 65.2\%$

If A and B have the same polydispersity index, then:

$$M_{w,A} = \text{PDI} \times M_{n,A}$$

$$M_{w,B} = \text{PDI} \times M_{n,B}$$

The weight-average molecular weight of the original mixture is:

$$M_{w,P} = w_A M_{w,A} + w_B M_{w,B}$$

Substituting the common PDI:

$$M_{w,P} = \text{PDI}(w_A M_{n,A} + w_B M_{n,B})$$

Therefore:

$$\text{PDI} = \frac{334603.93}{(0.3484)(128000) + (0.6516)(225000)}$$

$\text{PDI} = 1.75$
